

Manga109Script Dataset for Script-to-Layout Generation

Anonymous WACV Applications Track submission

Paper ID 967

Abstract

001 A well-designed layout plays a crucial role in visual me-
002 dia, serving to both capture the audience’s attention and
003 facilitate information transfer. While this principle applies
004 across various media, manga has a unique layout that re-
005 flects its narrative structure and is now a widely applied
006 medium in fields beyond entertainment, such as education
007 and advertising. In manga production, layouts are crafted
008 based on a narrative script; consequently, the layout is a
009 key component of visual direction, where the arrangement
010 of panels and characters is constructed to effectively convey
011 the narrative and enhance dramatic effect. Although layout
012 generation has seen progress in domains such as Web UIs
013 and posters that prioritize visual clarity, there has been lit-
014 tle exploration into script-based layout generation, largely
015 due to the lack of data. In this study, we created script data
016 for approximately 20,000 pages from the Manga109 dataset
017 and built the Manga109Script dataset linking scripts to
018 manga layouts. Leveraging this dataset, we developed
019 manga layout generation models that directly take scripts
020 as input, employing various LLMs as base models. Our
021 comprehensive evaluation shows that incorporating script
022 information demonstrably improves layout generation qual-
023 ity compared to existing methods. These results establish an
024 effective initial baseline for future research in script-based
025 layout generation, which we term Script2Layout. As such,
026 Script2Layout stands as a key enabling technology for ap-
027 plications designed to translate narratives into visual me-
028 dia, as exemplified by manga.

029 1. Introduction

030 Storytelling takes many forms, from text-focused novels to
031 visually-driven animation. Among these, manga has devel-
032 oped a unique expressive form that divides a flat surface into
033 panels containing characters and dialogue. In typical manga
034 production, the process of translating a narrative into a vi-
035 sual layout occurs in a preliminary draft called a “name.”
036 The layout therefore embodies critical directorial decisions.

037 Its composition, particularly the positioning of panels, char-
038 acters, and speech bubbles, is a deliberate choice designed
039 to structure the visual narrative and ensure the story is com-
040 municated clearly and with intended emotional weight [8].

041 In contrast to this narrative-centric process, the field of
042 layout generation has seen development, which has primar-
043 ily focused on functional applications like Web UIs [20] and
044 posters [9, 40], aiming to generate visually compelling lay-
045 outs that ensure clear and efficient information transfer [37].
046 However, layouts that combine storytelling and dramatic
047 effects, like manga, are under-researched. The absence of
048 data linking stories to layouts is a key reason for this gap.
049 Script data are often produced internally and rarely released
050 publicly, making it difficult to obtain. This data scarcity has
051 been the primary barrier to developing assistive applications
052 for this narrative-driven creative domain.

053 In this study, we utilized Vision Large Language Models
054 (VLLMs) to create Manga109Script, a script dataset for ap-
055 proximately 20,000 manga pages based on the Manga109
056 dataset [1, 28]¹. Recent advances in VLLMs have enabled
057 high-precision extraction of textual information from im-
058 ages [49]. This capability has significantly contributed to
059 research in Visual Storytelling, which extracts narratives
060 from image sequences [26, 30]. Despite advancements in
061 VLLMs, understanding content across multiple panels re-
062 mains challenging [15, 41]. To support VLLMs limita-
063 tions, we applied Visual Prompting [38] with the Manga109
064 dataset annotations and employed a rule-based algorithm
065 designed for panel reading order [34], to generate high-
066 quality narrative script data.

067 Our scripts follow standard script formatting conven-
068 tions, including *scene headings* indicating location, *action*
069 *descriptions* detailing character movements and scenery,
070 and *dialogue* (Fig. 1). We consulted with manga experts
071 to confirm the format fits professional workflows. The gen-
072 erated script’s quality was also evaluated by humans.

073 To demonstrate the practical utility of our dataset for
074 such creative applications, we developed manga layout gen-
075 eration models that take scripts as direct input. We com-

¹<http://www.manga109.org/en>

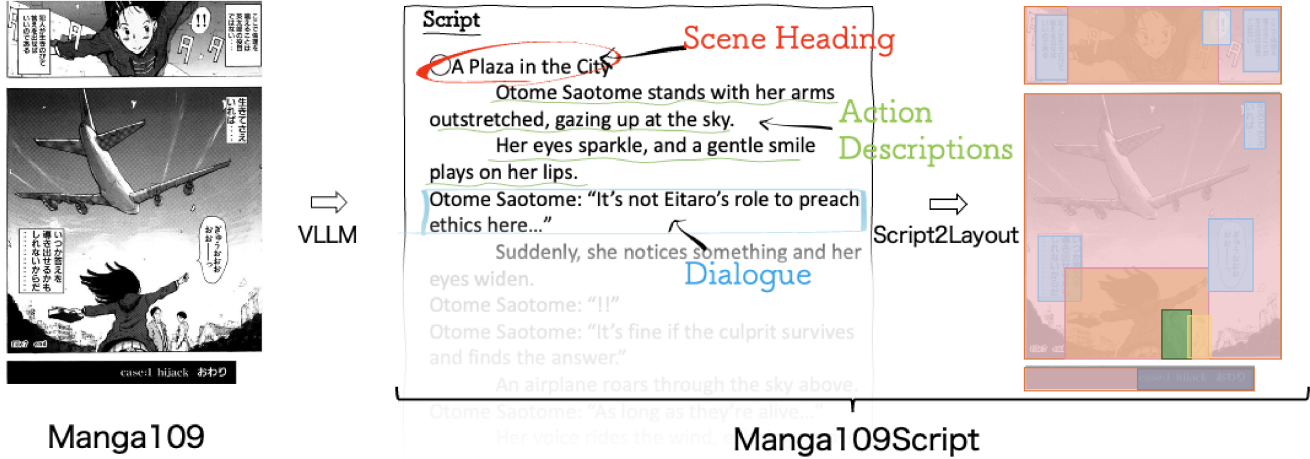


Figure 1. Overview of the Script2Layout task. From left to right: (1) a manga page from the Manga109 dataset; (2) its corresponding script, generated using a vision large language model (VLLM) with visual prompting and speaker-aware dialogue sequencing; and (3) the layout predicted by a Script2Layout model based on the generated script. The script follows standard manga production conventions and includes scene headings, action descriptions, and dialogue. This setup allows layout models to make use of narrative structure for generating more coherent and expressive layouts. (Source: *Hanzai Kousyounin Minegishi Eitarou* by Takashi Kii)

prehensively evaluated these models against existing layout generation methods and assessed the impact of script input to establish a baseline for the Script2Layout task. Dataset, and code access will be provided upon publication.

Our contribution is three-fold:

1. We introduce an annotation-assisted method to generate script data from manga images using VLLMs, validated by human evaluators.
2. We publish the first professional script-format manga storytelling dataset, covering 20k pages.
3. We establish robust baselines for the novel Script2Layout task, demonstrating for the first time how script influences manga layout.

2. Related work

2.1. Layout Generation

Layouts crucially impact viewers' comprehension and perception of visual content. The pursuit of excellent layouts across various media has driven research in layout generation models [13, 36, 37]. Previous research primarily targets layouts emphasizing element visibility, such as mobile UIs [7, 17], posters [9, 35, 40, 46], and PDFs [17, 50].

To meet the demands of practical applications and diverse design workflows, various constraint-based generation approaches have been developed. These approaches include generation conditioned solely on element types [18, 23], on both element types and size specifications [18, 22], and relational constraints between elements [21, 23]. There are also interactive variations, such as layout completion that infers missing elements from partial layouts [10], and layout refinement that improves existing layouts [33]. Layout generation with fixed background images

as a condition is called content-aware layout generation, which considers harmony with the background [12, 35].

Model development under these various constraints has enabled more accurate layout generation. Layout generation models have evolved from early GANs [24] and VAEs [3, 19] to Transformers [18] and diffusion models [4, 16, 48]. Recent studies have also explored the use of LLMs for this task [27, 40].

Despite these technical advancements, generating layout from narrative context is still unexplored. This challenge arises from dataset scarcity and the difficulty of converting narrative dynamics into visual layouts. Modern LLMs are now capable of interpreting complex contexts and nuances effectively [31]. Leveraging these capabilities, we propose Script2Layout, a task to generate layouts directly from scripts. We evaluate our approach against state-of-the-art methods such as LayoutDM[16], LayoutFormer++[18], and LayoutPrompter[27], thereby demonstrating both its effectiveness and innovative potential.

2.2. Manga Understanding as Visual Storytelling

High-quality script data is essential for the Script2Layout task, and its construction requires advanced techniques for narrative extraction from images.

Visual storytelling, the task of generating narratives from image sequences, was pioneered by Huang [14]. The field evolved from early models using Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) [47] to later methods using hierarchical attention [29], scene graphs [43], and causal reasoning [39]. These techniques move beyond simple image recognition to capture inter-image relationships and ensure narrative coherence.

Recent advances in VLLMs like GPT-4o [44] and Gem-

ini [2] have enhanced detailed context extraction from images, creating new opportunities in Visual Storytelling [45]. However, applying VLLMs to manga reduces performance due to a domain gap from its distinctive visual language, e.g., black-and-white art, unique reading orders, and discrete time and space within panels. These distinctive features make manga understanding a complex task composed of multiple intertwined subtasks. As illustrated in the CoMix benchmark [42], these include panel segmentation, dialogue ordering, and character re-identification across panels [34]. Character re-identification is particularly challenging, with GPT-4 achieving less than 50% accuracy [42].

To circumvent these challenges, we leverage solutions to key subtasks, providing the model with pre-processed inputs like ordered dialogue with speaker attribution [25] and annotated character regions [38]. This design allows the model to bypass low-level visual parsing and focus on generating high-quality narrative scripts, leveraging its core vision-language capabilities.

While datasets like Manga109Story [5] provide prose narratives, our primary aim is a *script-format* dataset. Accordingly, we developed Manga109Script with professional creators to fit conventional manga production workflows.

3. Manga109Script

Manga109Script is a script dataset corresponding to 19,555 manga pages from the Manga109 dataset [1, 28]. Manga109 contains manga images from 109 commercial Japanese manga works along with layout information such as panel, character, and text positions, made available for academic research with proper creator permissions. Manga109Script reformats the two-page spread layout information from Manga109 into single-page units, linking each page to its corresponding script data.

The script format follows a style commonly used in Japanese manga production. Each script consists of three main components: (1) Scene Headings, marked with a leading “○” to denote location, (2) Action Descriptions, written with a three-character indent to describe characters’ actions and environmental details, and (3) Dialogue, where the speaker’s name is followed by their speech enclosed in quotation marks. These components are illustrated in the center of Figure 1.

We adopted this script format because manga has primarily developed within Japan, and all manga titles in this study were originally created through Japanese production processes. This ensures the dataset aligns with actual manga creation practices.

3.1. Data Construction Process

To construct Manga109Script, we employed GPT-4o via API to generate page-level scripts from manga images in the Manga109 dataset. To support the model’s understanding of

character identity and dialogue context, we introduced two key inputs: (1) visual prompting, where the face of each character in the image was enclosed with a uniquely colored circle [38], and (2) pre-extracted dialogue annotations, including the textual content of each speech balloon and its associated speaker’s name [25]. The ordering of dialogues was determined in three steps: First, each dialogue balloon was assigned to the panel with which it had the highest Intersection over Union (IoU). Second, the reading order of panels on the page was determined using the algorithm proposed by Sachdeva *et al.*[34]. Finally, within each panel, dialogues were ordered based on their proximity to the top-right corner of the panel’s bounding box, following previous works [11, 34, 41].

To help the model recognize continuous narratives across sequential pages within the constraint of limited token length, we structured our interaction with GPT-4o into the following three components for each target page i :

- $\alpha(i)$: **Initial comprehension prompting**

We provided the full dialogue from the episode that includes page i and simulated an affirmative response from the model to promote narrative understanding.

Input: entire dialogue of the episode

Task: comprehend the episode

Agent: comprehension complete

- $\beta(i)$: **Script generation for the i -th page**

We provided GPT-4o with the processed image of page i (with visual prompting), the dialogue on page i , and the summary generated in step $\gamma(i-1)$, covering the story up to page $i-1$. The model was then instructed to generate the script for page i in the Japanese screenwriting format.

Input: page i image and dialogue,
and prior summary from $\gamma(i-1)$

Task: generate script for page i

Agent: script generation complete

- $\gamma(i)$: **Summarization up to the i -th page**

We then prompted the model to summarize the story up to page i to generate the summary input for $\beta(i+1)$.

Task: based on interaction history,
summarize the story so far

Agent: summarization complete

To generate the script for page i , we executed the sequence:

$$\alpha(i) \rightarrow \beta(i-1) \rightarrow \beta(i)$$

Here, the arrow ($\rightarrow$) indicates the chronological order in which each prompt was issued to GPT-4o.

To generate the summary of the story up to page i , we extended the sequence as follows:

$$\alpha(i) \rightarrow \beta(i-1) \rightarrow \beta(i) \rightarrow \gamma(i)$$

Table 1. Statistics of Manga109Script. "Distinct character types" is the count of unique characters identified per page, while "unique speaker types" is the number of distinct speaker IDs per page.

Element	Count	Avg / Page
<i>Layout Elements</i>		
Pages	19,555	—
Panels	103,850	5.31
Character instances	157,865	8.70
Distinct character types	60,812	3.11
Dialogue balloons	146,549	7.49
Unique speaker types	55,778	2.85
<i>Script Elements</i>		
Scene headings	26,931	1.38
Action descriptions	138,890	7.04
Dialogues	134,901	7.10
Japanese characters (text)	6,703,063	342.78

Note that for $i = 1$, these sequences become $\alpha(1) \rightarrow \beta(1)$ and $\alpha(1) \rightarrow \beta(1) \rightarrow \gamma(1)$, where $\beta(1)$ does not include a summary input.

By iteratively applying this series of interactions from $i = 1$ to the final page of each manga title, we constructed the full Manga109Script dataset. The dataset statistics are summarized in Table 1.

3.2. Quality Verification

Unlike general image captioning datasets, our scripts lack ground truth. Moreover, the script format provides contextual coherence rather than word overlap. Thus, we used human evaluation with specific metrics to verify script quality.

Evaluation items included: (1) *Scene Heading Recall*, determining whether locations depicted in the manga were correctly identified in script scene headings; (2) *Scene Heading Precision*, determining whether locations mentioned in script scene headings actually appeared in the manga; (3) *Action Description Precision*, determining whether action descriptions generated for a target page actually match that page when mixed with randomly selected action descriptions from other pages; (4) *Out-of-Context Action Description Detection Accuracy*, determining whether action descriptions from other pages could be correctly identified as not belonging to the target page; and (5) *Dialogue Recall*, determining whether dialogue appearing in the manga was accurately transcribed in the script.

We randomly selected approximately 10 consecutive pages from each work to maintain story continuity, totaling 1,000 pages. The 1,000 pages were divided into two sets of 500 pages each, with six evaluators each reviewing one set, resulting in three evaluations per page. These evaluators received fair compensation.

We calculated each evaluation metric by averaging correct judgments across all elements: Scene Heading Recall

(0.793), Scene Heading Precision (0.838), Action Description Precision (0.664), Out-of-Context Action Description Detection Accuracy (0.910), and Dialogue Recall (0.988).

4. Experiments

The primary goal of our experiments is to evaluate the effectiveness of using scripts as input for manga layout generation and to establish a baseline for Script2Layout. Using Manga109Script, we compare Script2Layout models (LLMs fine-tuned to generate layout coordinates conditioned on script input) with existing methods [16, 18, 27] under multiple constraint settings, including the presence or absence of script input.

4.1. Problem Formulation

We formulate manga layout generation as the task of predicting spatial positions for a given list of element categories. A layout l consisting of $N \in \mathbb{N}$ elements is represented as $l = \{(c_1, \mathbf{b}_1), \dots, (c_N, \mathbf{b}_N)\}$, where c_i denotes the category of the i -th element and $\mathbf{b}_i$ is its corresponding bounding box. In our experiments, we consider three types of element categories: *characters*, *speech balloons*, and *panels*. Each bounding box $\mathbf{b}_i$ is quantized on a 32×32 grid. Given the number of elements N and their categories $\{c_i\}_{i=1}^N$, the task is to predict their spatial positions $\{\mathbf{b}_i\}_{i=1}^N$ on the manga page.

This formulation reflects real-world manga production workflows, where artists compose layouts (known as "name" in Japanese production) based on a predefined script that specifies the categories of elements to be placed on each page, prior to any visual content creation. This formulation also bridges a critical gap between traditional layout generation models, which cannot directly accept script input, and Script2Layout models, which are designed to condition layout prediction on narrative scripts.

Under this formulation, we first describe how existing layout generation methods are applied to this task. We then introduce our Script2Layout models, highlighting the differences in how they utilize script information.

Existing Methods. The input and output formats of conventional methods are illustrated in the bottom part of Figure 2. Conventional layout generation methods require element categories to be predefined. In our experiments, categories are based on the three main types (*characters*, *speech balloons*, and *panels*) and are further distinguished on a per-page basis. Specifically, we assign category as $c_i \in \{\text{Chara}_i \mid i = 1, \dots, 25\} \cup \{\text{Speech}_i \mid i = 0, \dots, 25\} \cup \{\text{Panel}\}$. Here, Chara_i denotes the i -th character instance on a page, Speech_i represents a speech balloon spoken by Chara_i (or narration when $i = 0$). The number 25 was chosen to provide a buffer, as the maximum number of distinct characters in a single page of Manga109Script

Script2Layout

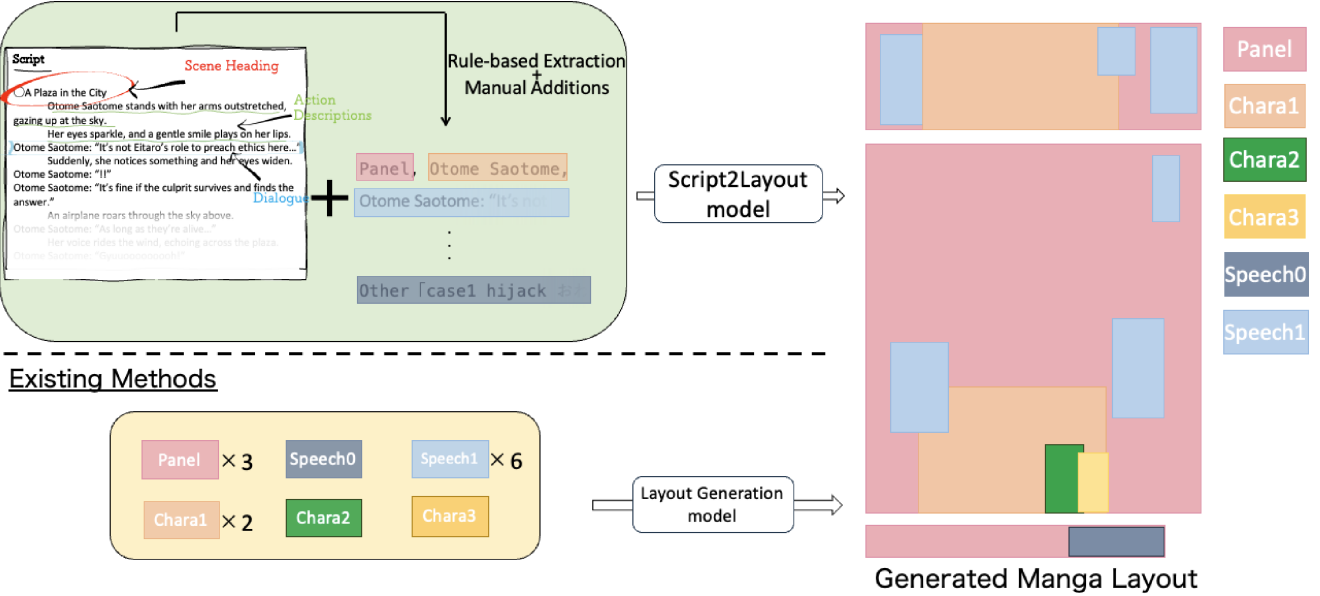


Figure 2. Comparison of manga layout generation approaches. Top: Our Script2Layout method takes natural language scripts and extracted layout elements (character names, dialogue, panels) as input to generate layouts via bounding box prediction. Bottom: Traditional methods require pre-defined category labels (e.g., Speech1, Chara2) as input. Unlike conventional approaches that rely on structured element types, Script2Layout enables direct layout generation from raw narrative scripts conditioned on storytelling intent.

is 22. Based on this formulation, we evaluate three representative layout generation methods: LayoutDM [16], LayoutFormer++ [18], and LayoutPrompter [27]. The first two models are trained on the manga layout task. LayoutPrompter, in contrast, is a prompt-based system that does not involve model training; we implemented it using the GPT-4o API. Although the system is capable of accepting script inputs, we did not provide script information in our experiments due to input length limitations. Consequently, LayoutPrompter was evaluated under the same conditions as other non-script-based methods. We adopt this generating conditioned on types (Gen-T), where the category of each element is known, as the base condition throughout this study. In addition, we evaluate the existing methods under two extended constraint settings:

- **Gen-TS (Generation conditioned on types and sizes):** Each element’s size (i.e., width and height) is given in addition to its category.
- **Gen-R (Generation conditioned on relationships):** Pairwise spatial relations (e.g., “element A is above element B”) between layout elements are provided as constraints.

Script2Layout Models. Under the above formulation, we use the following LLMs as base for Script2Layout models: T5 [32], Swallow-13B², Sarashina-13B³, and DeepSeek-R1-Distill-Qwen-14B-Japanese [6], selected for their strong

Japanese language capabilities and fine-tuning flexibility. The input and output formats of Script2Layout models are illustrated in the upper part of Figure 2. Unlike conventional methods that require predefined category labels, Script2Layout models can directly process linguistic content. They take as input the raw content of layout elements, such as character names and speech texts, extracted from the script. These contents are passed to the tokenizer of each LLM without assigning predefined category labels, allowing the model to interpret them directly based on its pre-trained knowledge. The model then outputs a quantized bounding box $\mathbf{b}_i = ((x_1, y_1), (x_2, y_2))$ for each element. In addition, we design input formats that allow the model to incorporate the script itself. Specifically, we consider the following three variants:

- **No Script:** No script information is included in the input.
- **Plain:** Raw script text is included as input.
- **Panel-Inserted:** Script input is augmented with explicit panel segmentation markers, indicating how the script content is divided across individual panels.

In addition, we further evaluated Script2Layout models under the Gen-TS setting, combining it with the **No Script** and **Plain** input conditions. This additional experiment aims to assess the extent to which script information contributes to layout prediction when the sizes of all elements are provided in advance.

²<https://swallow-llm.github.io/swallow-llama.ja.html>

³<https://huggingface.co/sbintuitions/sarashina13b>

4.2. Dataset Split

For training and evaluation, we filtered out 90 pages from Manga109Script that contained more than 50 layout elements, as such cases can cause combinatorial explosion in constraint-based models (particularly under Gen-R). The remaining 19,465 pages were used in our experiments.

We split the data by manga title to avoid title-level leakage across splits. The split maintains an 8:1:1 ratio, with 15,333 pages for training, 2,007 for validation, and 2,125 for testing. The training set was used to fine-tune LLMs for Script2Layout and to train conventional baseline models from randomly initialized parameters. The validation set was used for early stopping, and the test set for final performance evaluation.

4.3. Evaluation Metrics

We evaluate layout generation performance using two widely adopted metrics in the layout generation literature: Fréchet Inception Distance (FID) and mean Intersection over Union (mIoU). These metrics collectively assess both the fidelity and diversity of the generated layouts with respect to ground truth.

FID FID measures the similarity between distributions of real and generated layouts in a learned feature space. Following LayoutDM [16], we use FIDNetV3 as the feature extractor, which has been shown to be better suited for layout features than traditional image-based FID backbones. For Script2Layout, FID is computed between generated layouts and real layouts in the test set. In addition, as a reference, we also compute the FID between layouts in the validation and test sets and report it as “Real data.”

mIoU We compute mean Intersection over Union (mIoU) by first greedily matching generated and ground-truth elements based on maximum IoU per category. This approach, also used in prior works [16], accounts for category-specific spatial accuracy and is well-suited to our task setup where the element categories are known or inferred. For each test sample, the mIoU is calculated as the average over all matched pairs of elements, and the final score is averaged over the entire test set.

These two metrics complement each other: FID captures global distributional similarity and diversity, while mIoU reflects instance-level alignment between predicted and ground-truth layouts.

Qualitative Evaluation. In addition to quantitative metrics, we also conduct a qualitative comparison by visualizing generated layouts for selected pages in the test set. For each target page, layouts generated by different models are compared side-by-side against the ground-truth layout. This allows us to visually assess how well each model captures

spatial structure and how effectively script information contributes to layout coherence and accuracy.

4.4. Implementation Details

Our Script2Layout models were fine-tuned using the Low-Rank Adaptation (LoRA) technique. We utilized the Hugging Face `transformers` and `peft` libraries for all implementations. All training and inference were conducted on a single NVIDIA H100 GPU. A comprehensive list of hyper parameters, including the LoRA configuration and training parameters, is provided in the Appendix.

4.5. Evaluations and Discussions

Quantitative Results under Gen-T

Table 2 shows the FID and mIoU results for each model under the base constraint (Gen-T) across three script input conditions: No Script, Plain, and Panel-Inserted.

The FID results show that Plain script input significantly improves performance. Swallow-13B achieves the best FID score (18.415) under Plain despite poor No Script performance (35.845), demonstrating that even underperforming models can benefit from script information. Similarly, T5 and DeepSeek-R1 show consistent improvements from No Script to Plain, with three of four Script2Layout models benefiting from direct script input. These results support the effectiveness of Plain script input.

In contrast, Sarashina-13B achieves the best No Script performance (20.377) but degrades with script input (24.191 for Plain, 25.127 for Panel-Inserted). This suggests that narrative input may introduce noise for models already performing well with raw content alone.

Panel-Inserted conditions only improve DeepSeek-R1 (21.094), while other models perform worse than Plain. This suggests that panel-wise segmentation cues requires strong reasoning capabilities to leverage effectively.

Among existing methods, LayoutDM achieves the best FID (22.428). While LayoutDM previously scored 7.95 on PubLayNet [50] where real data baseline is 6.25 [16], the increased FID reflects manga layout generation’s unique difficulty. Given this challenging, four model-condition pairs surpass LayoutDM: Sarashina-13B (No Script), Swallow-13B (Plain, Panel-Inserted), and DeepSeek-R1 (Panel-Inserted), with three cases using script input, demonstrating script effectiveness.

The advantage of our approach is even more evident in the mIoU scores. All Script2Layout models not only improved with script input but also, with one exception, outperformed all existing methods. This strongly suggests that generating layouts from semantic content, rather than abstract categories, provides a decisive advantage in capturing fine-grained spatial relationships.

Table 2. FID and mIoU scores under the Gen-T setting. Script2Layout models are evaluated with three types of script input. Most models outperform existing baselines, especially in mIoU, demonstrating the effectiveness of script-based guidance.

Model	Script Type	No Script		Plain		Panel-Inserted	
	Metric	FID ↓	mIoU ↑	FID ↓	mIoU ↑	FID ↓	mIoU ↑
<i>Layout Generation Models</i>							
LayoutDM		22.428	0.161	-	-	-	-
LayoutFormer++		28.402	0.140	-	-	-	-
LayoutPrompter		133.502	0.138	-	-	-	-
<i>Script2Layout Models</i>							
T5 (Fine-Tuned)		29.549	0.201	28.744	0.210	31.287	0.204
Swallow-13B (Fine-Tuned)		35.845	0.156	18.415	0.182	21.557	0.184
Sarashina-13B (Fine-Tuned)		20.377	0.192	24.191	0.207	25.127	0.213
DeepSeek-R1-Japanese-14B (Fine-Tuned)		24.618	0.174	22.502	0.186	21.094	0.184
Real data		4.780		4.780		4.780	

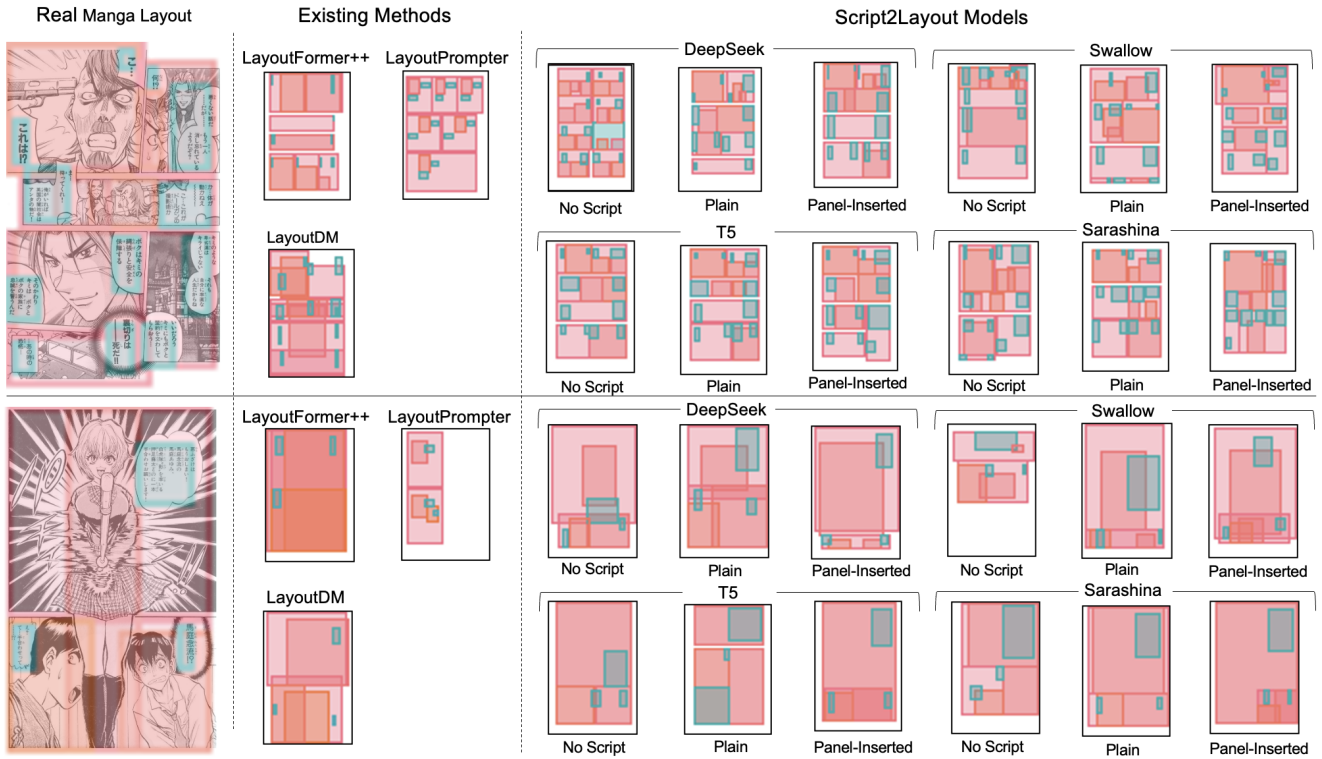


Figure 3. Qualitative comparison of generated layouts. Each row shows one example: the real manga layout (left), followed by predictions from existing methods and Script2Layout models with different script input settings (No Script, Plain, Panel-Inserted). Script2Layout models generally produce layouts closer to the real ones, particularly when script information is provided. (Source: *Doll Gun* by Ryusei Deguchi, *Touta Mairimasu* by Shinji Saijo)

Qualitative Results

Figure 3 visualizes generated layouts, showing how qualitative insights into how different models, especially Script2Layout models, respond to input.

In the top row, the target page features a dramatic close-up of a character pointing a gun at his own temple. Script2Layout models with script input consistently place large character regions at the top-left, mirroring the orig-

inal page structure. This alignment likely reflects textual cues such as "Norman stands frozen in shock. His eyes are wide open, with sweat beading on his forehead," which may have guided the models to allocate spatial prominence to the main character.

The bottom row shows a page with one dominant character with a bamboo sword. Script2Layout models with script input place a large character region centrally with a speech

Table 3. Evaluation results of existing layout generation methods under different constraint settings.

Model	Constraint	FID ↓	mIoU ↑
LayoutDM	Gen-T	22.428	0.161
	Gen-TS	20.514	0.271
	Gen-R	21.405	0.053
LayoutFormer++	Gen-T	28.402	0.140
	Gen-TS	17.487	0.208
	Gen-R	26.848	0.158
LayoutPrompter	Gen-T	133.503	0.138
	Gen-TS	58.993	0.148
	Gen-R	130.502	0.097

balloon at the upper right, mimicking the original composition. The script line "Ayumi Maniwa holds her bamboo sword at the ready, standing with a dignified posture" likely encouraged this semantically meaningful layout.

Overall, these examples suggest that leveraging script information is a promising approach for generating layouts with greater narrative coherence and expressiveness.

Constraint Analysis

We first evaluated the impact of additional geometric constraints. Relational constraints (Gen-R), defining explicit pairwise element relationships, offered little to no performance gain over the Gen-T baseline for conventional models (Table 3). This suggests that simple pairwise relations are less effective than the rich, contextual guidance provided by a narrative script.

In stark contrast, providing element sizes (Gen-TS) dramatically boosted performance across all models, both conventional and our own (Table 3 and 4). For instance, LayoutDM's mIoU jumped from 0.161 to 0.271, while our T5 model reached 0.394, affirming the critical role of size information. Crucially, even with size information provided, adding the script further improved performance for our models. This demonstrates the complementary nature of geometric constraints (size) and semantic context (script), indicating that integrating both is a key direction for future work.

Failure Cases and Limitations

Despite the promising results, our approach has several limitations. First, its reliance on bounding boxes prevents it from capturing the full geometric complexity of manga, such as slanted or non-rectangular panels. Second, because our models directly process the script's textual content, they can generate disproportionately large speech balloons for long dialogues (Figure 4). Finally, our LLM-based approach incurs significant computational cost; processing scripts results in in per-sample generation time of approximately 7-12 seconds depending on script length slower than

Table 4. Performance of Script2Layout models under Gen-T and Gen-TS constraints, with and without script input.

Model	Script Type	No Script		Plain	
		FID ↓	mIoU ↑	FID ↓	mIoU ↑
T5	Gen-T	29.549	0.201	28.744	0.210
	Gen-TS	19.445	0.379	19.734	0.394
Swallow	Gen-T	35.845	0.156	18.415	0.182
	Gen-TS	20.650	0.337	19.946	0.334
Sarashina	Gen-T	20.377	0.192	24.191	0.207
	Gen-TS	18.165	0.358	18.783	0.352
DeepSeek	Gen-T	24.618	0.174	22.502	0.186
	Gen-TS	20.769	0.381	20.013	0.383

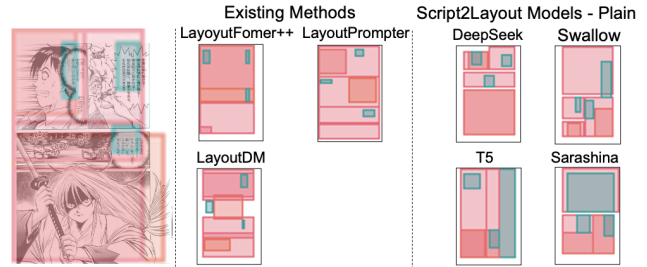


Figure 4. Failure cases of generated layouts. Each row shows examples: the real manga layout (left), followed by predictions from existing methods and Script2Layout models (Source: *Touta Mairimasu* by Shinji Saijo)

baselines like LayoutDM (approx. 0.66s). Addressing these challenges, future work could also explore conditioning diffusion models on scripts to mitigate these limitations, particularly in efficiency.

5. Conclusion

In this work, we introduced Script2Layout, a framework for manga layout generation from scripts, and constructed Manga109Script, the first large-scale dataset to enable this line of research. Our experiments demonstrate that conditioning on script provides a clear advantage over existing methods, particularly in mIoU. This work provides valuable resources for manga production and demonstrates broader applicability to narrative visualization systems. Future research directions include exploring the complementarity between size and script information, investigating more suitable base models, and evaluating practical system implementations. These advances will contribute to the field of story-to-visual content generation applications.

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